

School of Computing & IT
Web Programming (CIT2011)
Week 1 – Exploring the Web with Developer Tools

1. Lab Overview

Modern web developers need to understand what happens inside the browser when a webpage loads and when a user interacts with it. In this lab, students use Chrome, or Firefox Developer Tools to inspect HTML, and CSS.

Register to <https://onecompiler.com/>. You will use this online compiler to share your ongoing labs with your tutor for review.

2. Learning Outcomes

1. Open and navigate browser Developer Tools.
2. Identify HTML elements using the Elements/Inspector panel.
3. Modify HTML and CSS temporarily using Developer Tools.
4. Use responsive/device emulation to test different screen sizes.
5. Use Developer Tools as part of a modern web-development workflow.

3. Software Requirements

Students need the following:

- Google Chrome / Mozilla Firefox
- Internet connection
- A text editor such as VS Code

4. Developer Tools Overview

Chrome	Firefox	Purpose
Elements	Inspector	Examine HTML and CSS
Console	Console	Execute JavaScript and view errors
Sources	Debugger	Examine JavaScript/source files
Network	Network	Examine HTTP requests
Application	Storage	Examine cookies/storage
Lighthouse	Lighthouse	Performance/accessibility analysis
Device Toolbar	Responsive Design Mode	Test different screen sizes

PART A — Opening Developer Tools

Activity 1: Open Developer Tools

- a. Open <https://www.w3schools.com/html/default.asp> or another publicly accessible website of your choice).
- b. Open Developer Tools using right-click → Inspect, F12, Ctrl+Shift+I on Windows, or Cmd+Option+I on macOS.

Question 1. What is the name of the panel used to inspect the webpage's HTML?

Elements

Question 2. List five panels available in Developer Tools.

1. Elements 2. Console 3. Sources
4. Network 5. Performance

PART B — Inspecting HTML

Activity 2: Examine the HTML Structure

Select the Elements/Inspector panel. Locate <html>, <head>, <body>, headings, paragraphs, links, images and navigation elements.

Question 3. Identify one heading element. Record the HTML tag and the text inside

it. HTML tag: <h2>

Text: Learning by examples

Question 4. Find one hyperlink. Record the HTML element and its href value.

HTML element: See all HTML Examples

href: html examples.asp

Activity 3: Modify HTML

Find a visible heading. Right-click the element and select Edit as HTML (or double-click it). Temporarily change it to:

<h1>CIT2011 Developer Tools Lab</h1>

Question 5. What happened to the webpage?

The heading on the webpage changed and updated instantly in the browser showing “CIT2011 Developer Tools Lab” instead of its original text.

Question 6. Was the original website permanently changed? Explain why.

No the original website was not permanently changed. The change was only made temporarily in the browser. If the page is refreshed the original content will appear again.

PART C — Inspecting CSS

Activity 4: Examine CSS

Select an HTML element containing visible text. Examine the Styles panel and identify CSS properties controlling font size, font family, margin, padding, text colour and background colour.

Property	Value for html example
font-size	32px

font-family	“Segoe UI”, Arial, sans-serif
margin	Margin block end - 10px, margin block start - 10px, margin bottom - 10px, margin in-line end - 0px, margin in-line start- 0px, margin left - 0px, margin right - 0px, margin top -10px
padding	0px
color	rgb(0, 0, 0)
background-color	

Activity 5: Modify CSS

Add the following declaration to a heading in the Styles panel:

font-size: 50px;

Then change it to:

font-size: 20px;

Question 1. What happened when the font-size value changed?

When I changed the font size to 50px the heading became much larger but when I changed it to 20px the text became smaller.

Question 2. Add text-transform: uppercase; What happened?

All the letters in the heading instantly changed to capital letters.

Question 3. Change margin and padding. Explain the difference between margin and padding.

Margin is the space outside an element border that creates distance between the element and other elements around it. When I increased the margin the heading was pushed away from the other content.

Padding is the space between the content and the border. When I increased the padding the background above the heading grew larger pushing the text further down but other elements barely moved.

PART D — Understanding the Box Model

Activity 6: Inspect the Box Model

Select an element and locate the Box Model diagram.

Question 4. Identify the four major components of the CSS box model.

1. Margin 2. Border 3. Padding 4. Content

PART I — Responsive Web Design

Activity 15: Device Emulation

Open Device Toolbar / Responsive Design Mode and test mobile, tablet, laptop and desktop sizes such as 375×667, 768×1024 and 1366×768.

Question 5. What happened to the webpage when the screen width became smaller?

When the screen width became smaller the webpage rearranged itself to fit the smaller screen which made the content for the overall page that was side by side became taller and more vertically organised.

Question 6. Did the navigation change? Describe what happened.

Yes the navigation changed. On the desktop view (1366 × 768) the top menu showed all links in a horizontal row but when I switched to mobile size (375 × 667) the navigation showed a three line icon that hides the links until it is clicked.

Question 7. Did images resize? Explain your observation.

Yes the images resized to fit within the smaller screen so they do not take up more space than the available screen width.

Activity 16: Test Your Own Responsive Design

Create a file named responsive.html using the following code:

```
<!DOCTYPE html>
<html>
<head>
<title>Responsive Test</title>
<style>
body { font-family: Arial; margin: 20px; }
.container { display: flex; gap: 20px; }
.box { padding: 30px; background-color: lightblue; flex: 1; }
@media (max-width: 600px) {
.container { flex-direction: column; }
}
</style>
</head>
<body>
```

```
<h1>Responsive Web Design</h1>
<div class="container">
<div class="box">Box 1</div>
<div class="box">Box 2</div>
<div class="box">Box 3</div>
</div>
</body>
</html>
```

Use Developer Tools to test 1200px, 900px, 700px, 600px, 500px and 375px.

Question 8. At approximately what width does the layout change from rows to

columns? The layout changes from rows to columns at 600px

Question 9. Which CSS feature caused the layout to change?

The CSS media query caused the layout to change. It tells the browser to switch row to column.

Question 10. Complete the table above.

FINAL CHALLENGE — Activity 20: Become the Web Detective

A student reports that a webpage is displaying incorrectly on their mobile phone, although it works correctly on a laptop. Use Developer Tools to investigate responsive layout, CSS rules, media queries, images, JavaScript errors, network requests and HTTP status codes.

Question 11. Identify three possible reasons why a webpage may work correctly on desktop but incorrectly on mobile.

1. Images or other elements may be too large because they were not resized properly for a mobile screen.
2. Large unoptimised images can cause a page to load slowly on mobile networks and if the connection times out images may fail to load completely
3. Font sizes may be too large or too small on mobile making the main content hard to read.

FINAL REFLECTION

Question 12. How does Developer Tools help a web developer understand a webpage?

Developer tools help a web developer understand a webpage by letting them to view the code of the page while it's running in the browser where they can inspect the html structure, see the css rules being applied and test how the page behaves on different screen sizes.

Question 13. Why is it useful to inspect HTML and CSS rather than simply looking at the webpage?

It is useful because looking at the webpage only shows the final result but when looking into the html and css the developer can see what the actual structure and styles on the webpage looks and works.

Question 14. Why is the Network panel important to a web developer?

The network panel is important because it shows all the requests the browser makes when loading a page which helps developers to identify any request that has failed, slow loading, broken API calls, or missing files when a page is not loading correctly.

Question 15. Why is responsive testing important?

Responsive testing is important because it allow a user to access websites on many different devices such as phones, tablets, laptops that have different screen sizes. Which ensures that a layout, text, images, and navigation all fit properly to every screen size so every visitor enjoy a user friendly experience regardless of their device.

Question 16. How can Developer Tools help identify bugs?

Developer tools can help developers identify bugs by inspecting the html and css, check for javascript errors within the console, and to examine the network requests to find problems with the webpage.

Question 17. Explain how HTML, CSS and JavaScript work together to create an interactive webpage. HTML provides the structure and content of the webpage. CSS control is responsible for the appearance and layout of the webpage. The javascript provides interaction which allows the website to interact with the user. Together they create an interactive webpage.

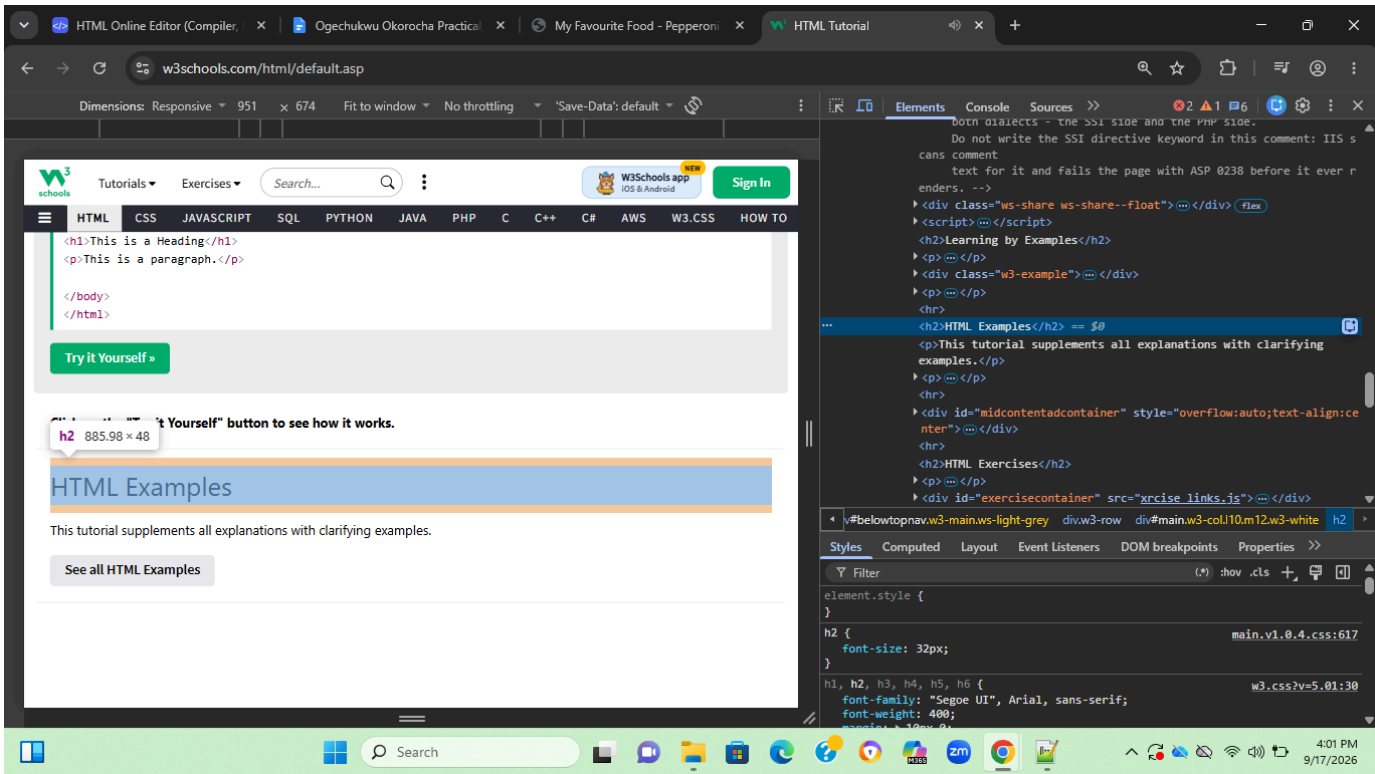
Question 18 – Build a webpage

Use the template provided by your tutor to build a webpage using HTML5 and basic CSS3. Webpage must include, basic HTML5 tags such as, image, paragraph, navigation, heading, sections, footer, links, etc.

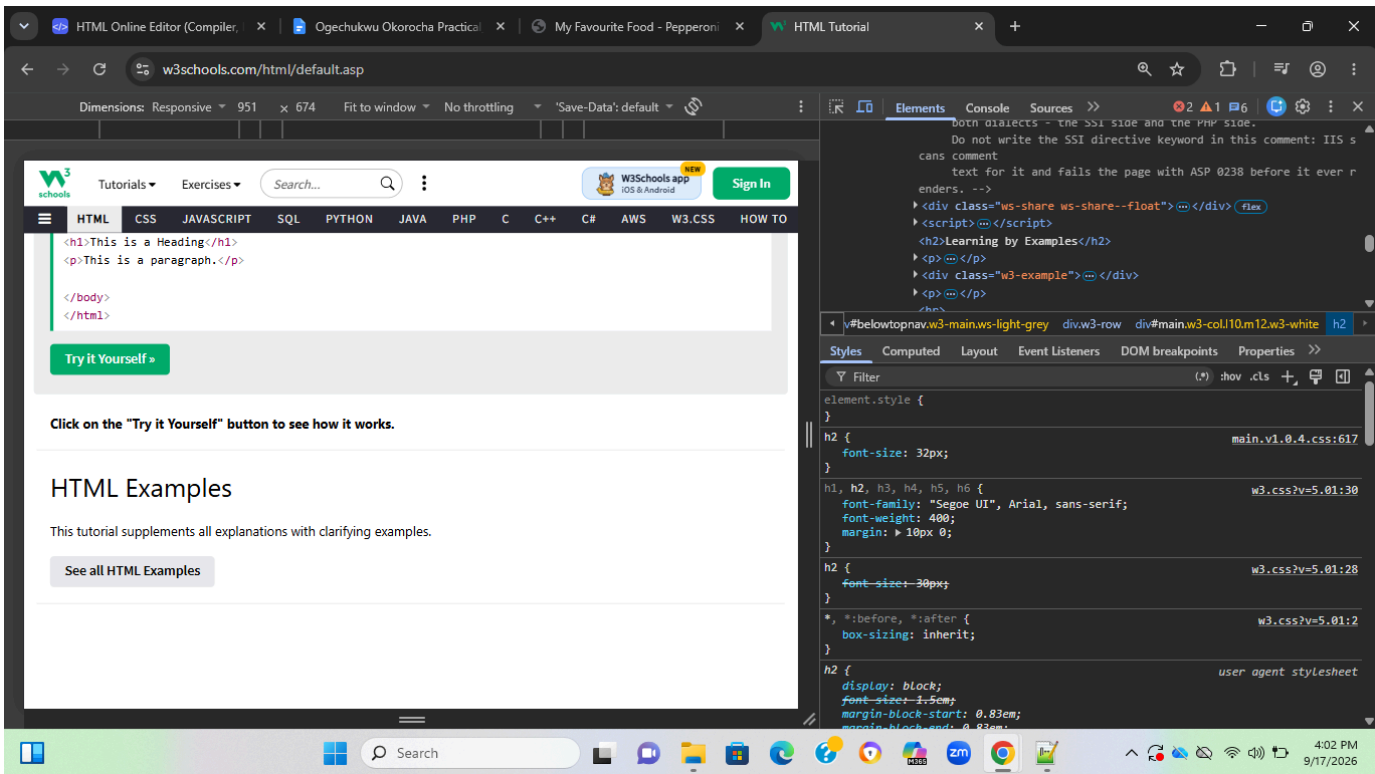
Practical Deliverables

- Completed lab sheet (Questions 1–18).
- At least six screenshots: HTML inspection, CSS inspection
- responsive.html

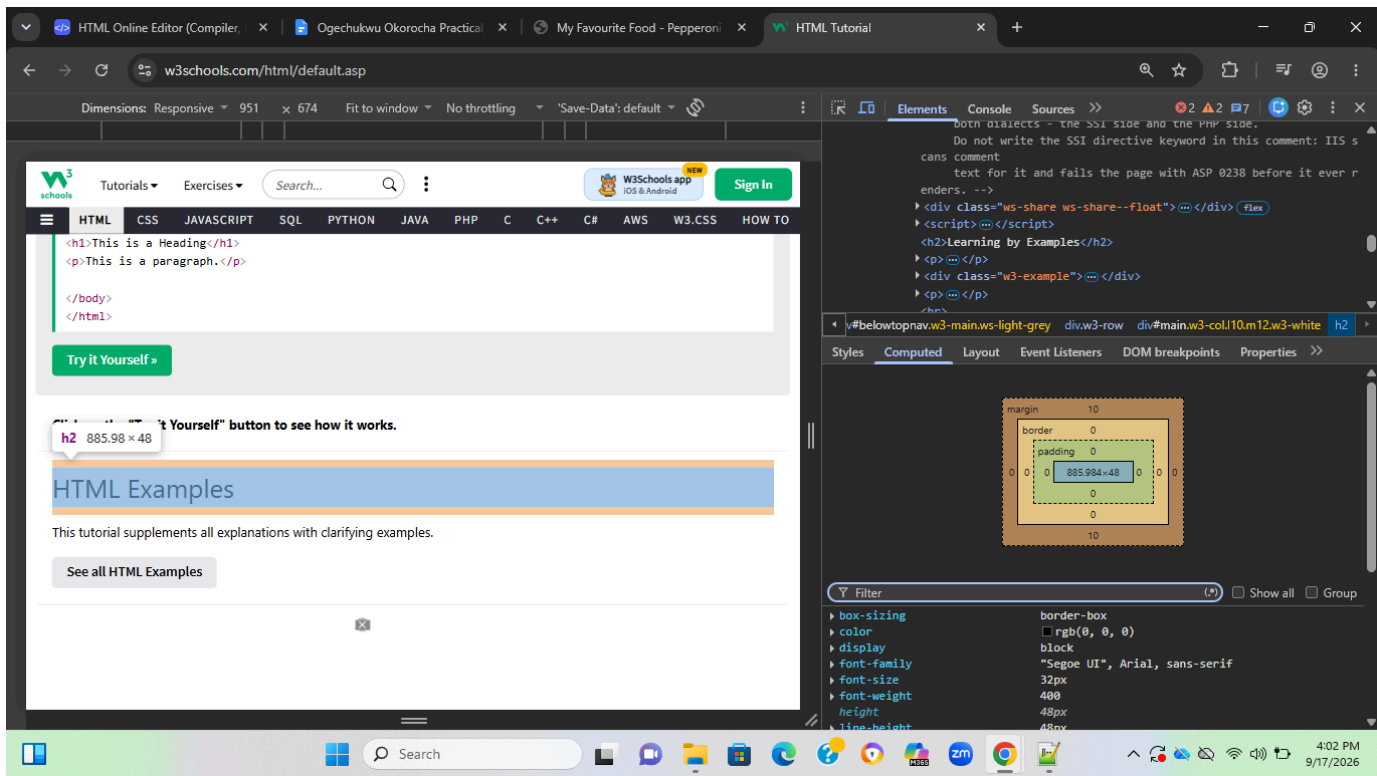
Screenshot 1 — HTML Inspection



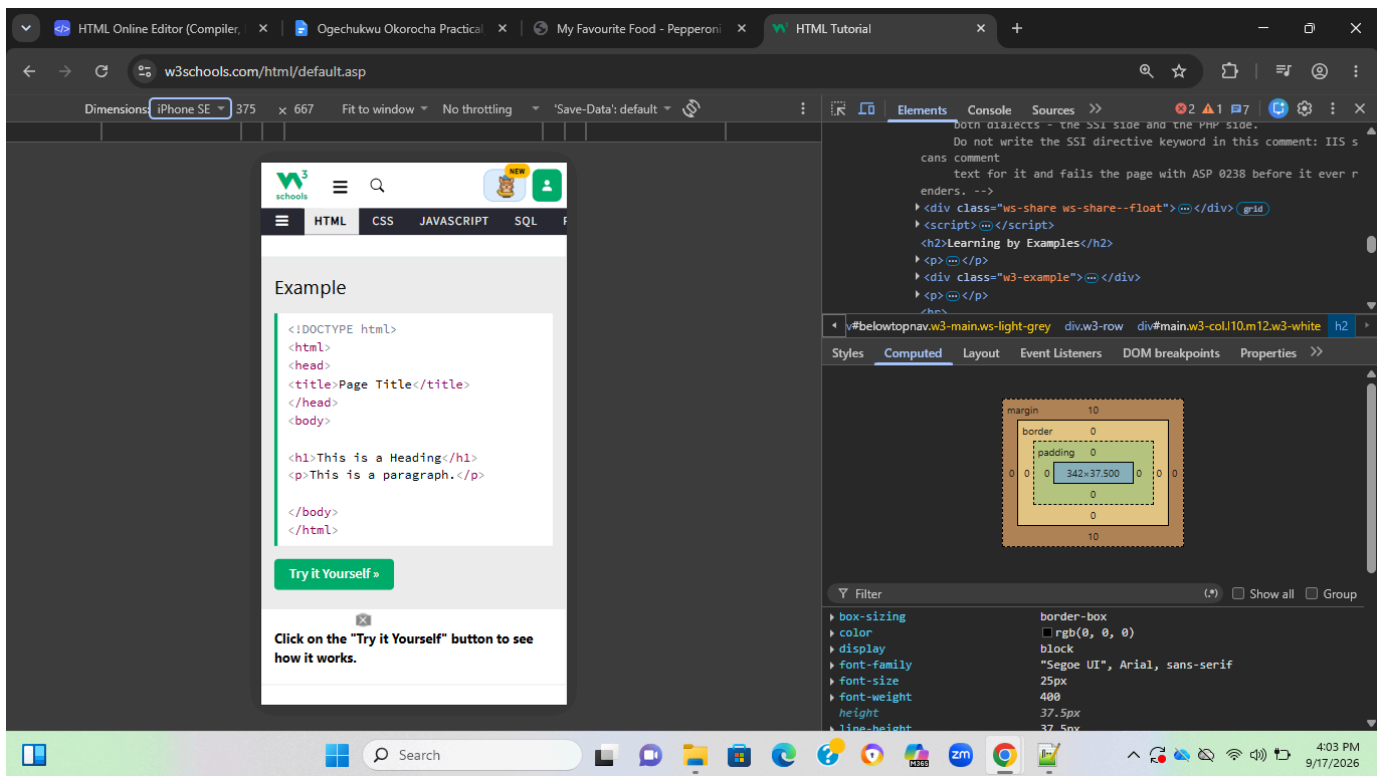
Screenshot 2 — CSS Inspection



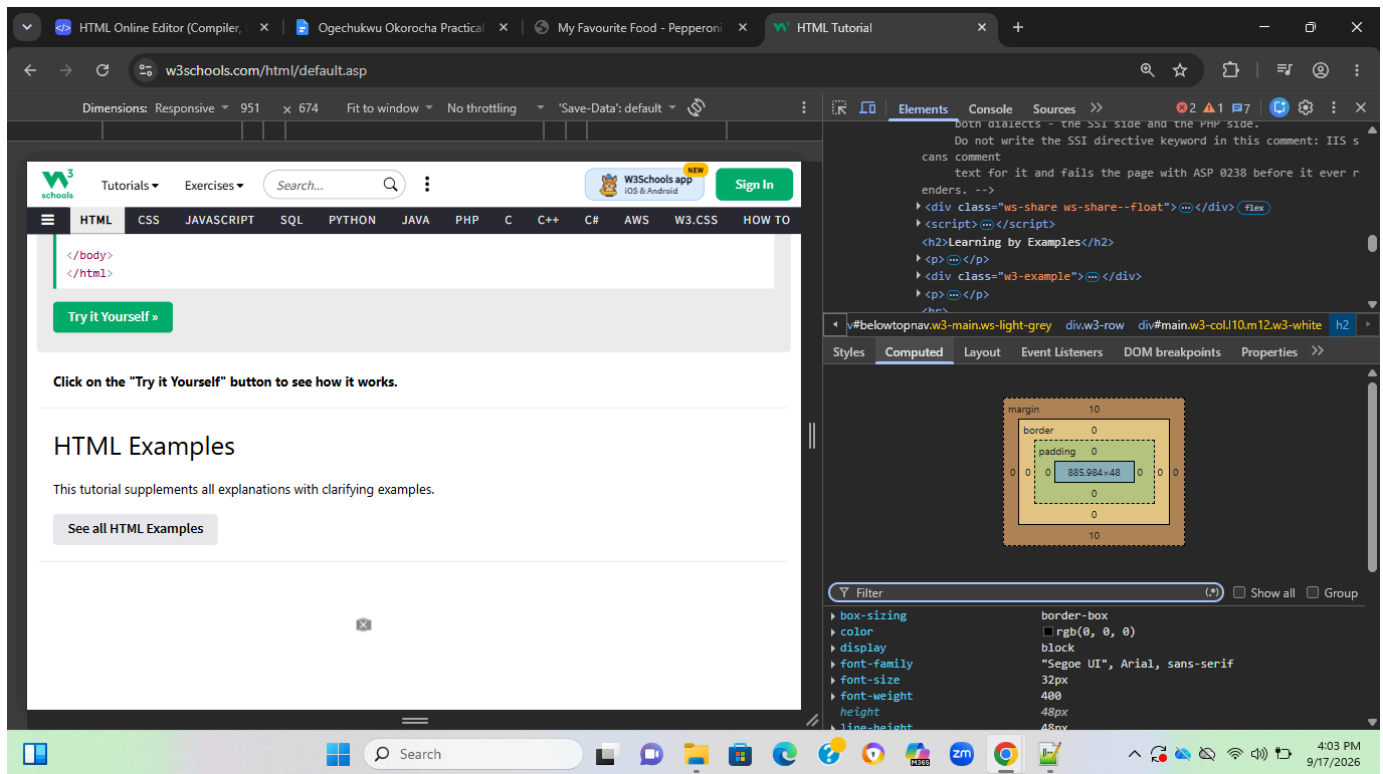
Screenshot 3 — Box Model



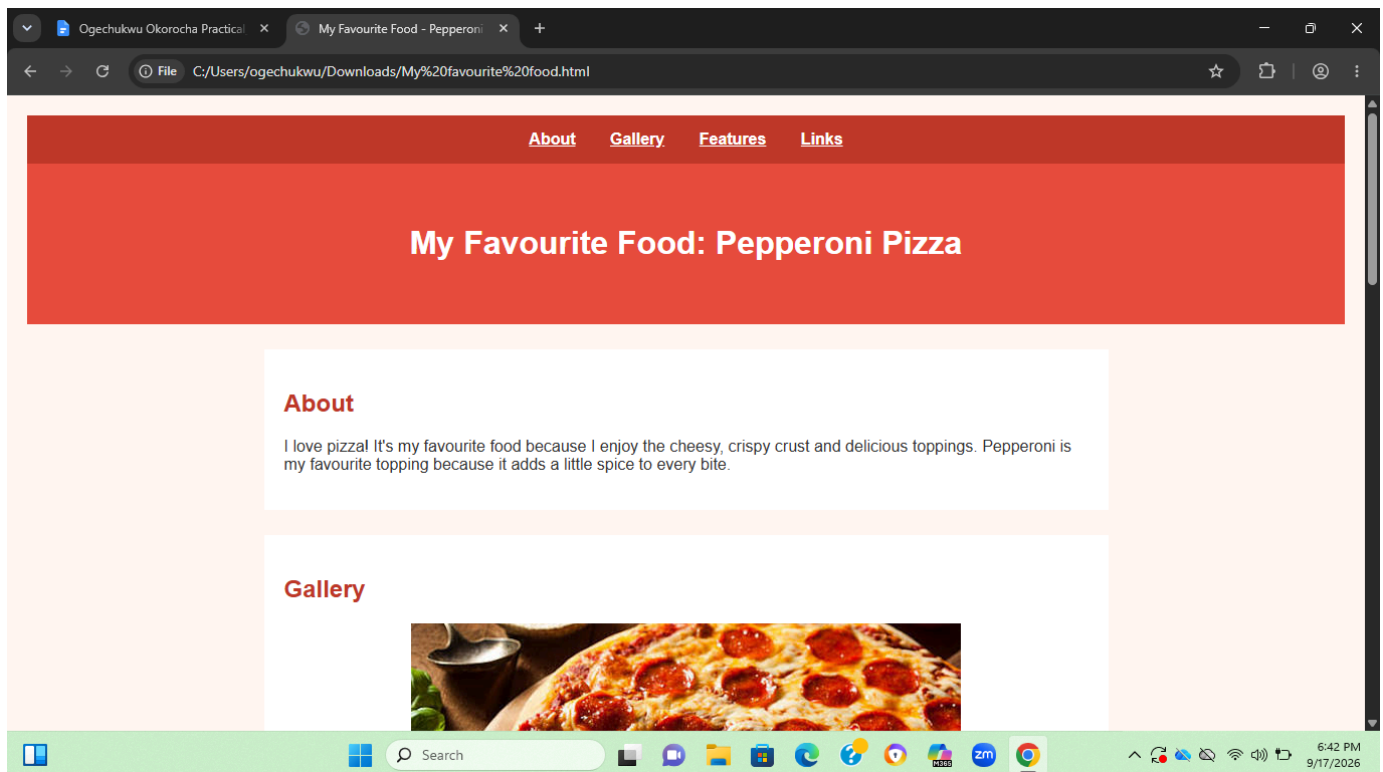
Screenshot 4 — Mobile View



Screenshot 5 — Desktop View



Screenshot 6 — Pizza Page



Link to my webpage: <https://onecompiler.com/html/453jfx2qt>